Krish Kochar

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Education

University of Waterloo Sep 2022 - Apr 2027

Bachelor of Computer Science (AI Specialization) ***GPA: 4.0***

Selected Coursework: Algorithms, Operating Systems, Concurrency, Distributed Systems, Machine Learning, Computer Security

Experience

Amazon Web Services (AWS) Sep 2026 - Dec 2026

Software Engineering Intern, Journal Service Control Plane

- Building a dynamic rate limiter for a critical Tier-0 service handling **100M+ QPS** across S3, DynamoDB, Kinesis & more

Atlassian May 2026 - Aug 2026

Software Engineering Intern, 3P Connector Data Ingestion

- Built & shipped a **Kotlin** resource attribution layer for Rovo AI's connector platform, providing the first per-connector visibility into **5M+ resource calls/min** across **DynamoDB**, **SQS**, **Redis**, & compute in production
- Designed non-blocking hot-path instrumentation with concurrent maps, atomics, & windowed aggregation; sampled **1-in-128** payloads to maintain an EWMA size estimate, avoiding per-call serialization
- Cut p50 instrumentation latency from **8ms to 1ms** after tracing ∼**98%** of wall time to a config lookup & removing it
- Used the resulting telemetry to trace **6M SQS calls/15min** to an abandoned Asana workflow left enabled for **14+ months**; disabled the stale feature flag, eliminating ∼**$160/day** in wasted staging spend

Super.com May 2025 - Aug 2025

Software Engineering Intern, Flights Backend

- Engineered **Go** backend services for booking, payment, & refund workflows, achieving **sub-50ms p90** service latency
- Designed an async webhook-based refund dispute pipeline with idempotency guarantees, ensuring correct semantics under retries, duplicate events, & partial failures, reducing **resolution turnaround time by 30%**
- Drove production triage, resolving **5+** critical incidents & introducing debugging workflows that reduced **MTTR by 50%**

OTTO by Rockwell Automation Jan 2025 – Apr 2025

Software Developer Intern, Fleet Core

- Developed a multi-fleet interoperability system in **C++**, **Python**, & **ROS2**, improving coordination across **500+** robots
- Redesigned **networking architecture** by migrating from CycloneDDS to Zenoh, reducing latency by **25%** & increasing throughput by **30%**, validated with benchmarks & a production-ready **PoC** for fleet deployment
- Replaced polling with an **event-driven ROS2** topic for real-time robot tracking, reducing CPU usage by **20%**

Ford May 2024 - Aug 2024

Platform Software Developer Intern, Vision Compute Service

- Deployed a real-time Vision Compute Service in **C++** with **OpenVX** & **OpenCV**, accelerating sensor data processing & optimizing multi-threaded workloads for high-performance execution on embedded hardware
- Integrated **Gcov** into **Jenkins** CI pipelines, enabling **80%** functional test coverage visibility & unblocking QA workflows
- Debugged & resolved **critical memory defects** with Valgrind, eliminating leaks & stabilizing production firmware

Research

University of Waterloo Jan 2026 - Apr 2026

Undergraduate Research Assistant, Prof. Sujaya Maiyya

- Built a **C++** oblivious **HNSW** vector-search prototype to reduce ANN access-pattern leakage in secure similarity queries
- Benchmarked **TEE**-based oblivious search, quantifying latency, memory overhead, & graph-traversal leakage across queries

University of Waterloo & Vector Institute Sep 2024 - Dec 2024

Undergraduate Research Assistant, Prof. Xi He

- Implemented & evaluated **membership inference attacks** on diffusion-generated synthetic tabular datasets with Python

Technical Skills

Languages: C, C++, Python, Go, Rust, Java, Kotlin, SQL, TypeScript, Scala, Bash

Systems & Tools: Docker, Kubernetes, Spark, Redis, Jenkins, Git, Datadog, Splunk, ROS2, OpenCV, PyTorch, FastAPI